



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Department of Statistical Sciences
"Paolo Fortunati"

Second Cycle Degree
M.Sc. Quantitative Finance

Trading Under Uncertainty

A Meta-Verification Framework for Machine
Learning in Intraday Forex Trading

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Outline

1. Motivation & Research Question
2. Framework & Methodology
3. Results: Tournament, Production & Experiment
4. Conclusion

The thesis in one sentence

The value of this ML trading pipeline in intraday FX lies less in predicting the market than in **deciding when *not* to trade**.

01

Motivation & Research Question

Quantifying the Unknown

- Intraday FX: **vanishingly low signal-to-noise ratio** and non-stationary regimes
- Given a random walk with drift, an ML model will *a/ways* find something. Is it structure, or a ghost in the data?
- The gullible approach treats price series as a standard supervised learning problem
- This thesis: shift from **point-estimate trading** to **meta-verification**

1 Data-Mining Trap

Exhaustive HPO on many features finds *stochastic flukes*, not alpha

2 Hard Thresholds

A classifier trades at $p_{\text{long}} = 50.01\%$ exactly like at 95%. Costs would erode the edge (if any).

3 Regime Blindness

One model for G10 majors *and* emerging markets ignores different liquidity and volatility.

Research Question & Objectives

Can we build an ML classifier that drives a Forex trading model successfully?

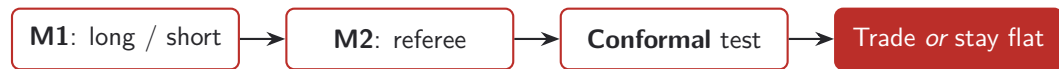
	Objective	Question
1	Architecture efficacy	Which of six ML architectures performs the least bad under purged, nested walk-forward validation?
2	Meta-machinery	Can M2 veto weak M1 trades, and does conformal-like calibration beat a fixed cut-off?
3	Discovery control	Can the deflated Sharpe ratio (DSR) separate luck from systematic alpha?
4	Feature anatomy	Which features give stable signals, and why?
5	Global vs. local	One global model with pair-specific guardrails. Does it work for Majors and EMs?

02

Framework & Methodology

The Meta-Verification Framework

Every signal must survive a chain of adversarial checks before it is traded



Triple-barrier labels

$$y_i = \begin{cases} 1 & \text{TP first} \\ -1 & \text{SL first} \\ \text{sgn}(r_i) & \text{time-out} \end{cases}$$

Path-dependent labels:

146,711 in total (52.7% short vs 47.3% long).

Meta-labels for M2

$$y_i^{\text{M2}} = \begin{cases} 1 & \hat{y}_i^{\text{M1}} = y_i \\ 0 & \hat{y}_i^{\text{M1}} \neq y_i \end{cases}$$

M1 picks the *side*, M2 decides *whether* to bet.

Conformal veto (H_0 : signal is a mistake)

$$p = \frac{|\{i : \alpha_i \geq \alpha_{\text{new}}\}| + 1}{k + 1}$$

$$\text{act} \iff p \leq \varepsilon$$

α = M2 score

calib. set = **past M1 mistakes only**

ε bounds the Type-I error (with a caveat).

FX data is not exchangeable (**order matters**) $\Rightarrow$ the veto is more like a disciplined heuristic whose calibration is *checked out-of-sample*.

An Effort to Guard Against False Discovery

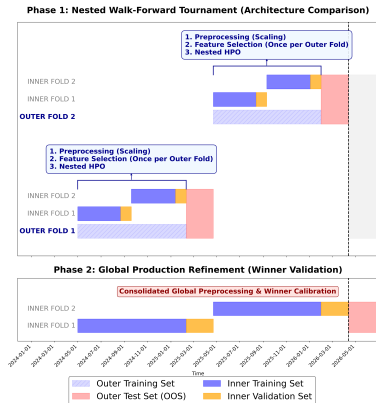
Purged, nested walk-forward

- 2 outer $\times$ 2 inner folds: preprocessing & feature selection once per outer fold
- Drop training obs. whose label window $[t_i, t_i + h_i]$ overlaps a test label
- Global test set touched **exactly once**, by the tournament winner

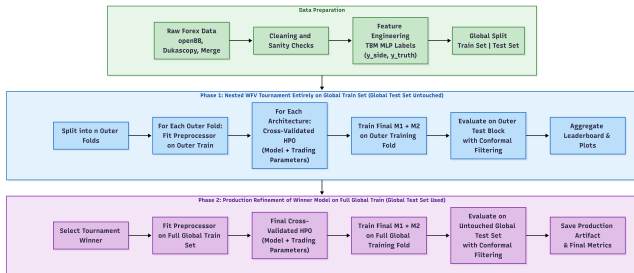
Deflated Sharpe ratio

$$DSR = \Phi \left[\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \gamma_3 \widehat{SR} + \frac{\gamma_4 - 1}{4} \widehat{SR}^2}} \right]$$

- SR^* = best Sharpe expected *by chance* after N trials ($N = 1,250$ per model & fold. 5,000 in production)
- $DSR < 50\% \Rightarrow$ indistinguishable from search luck



Research Pipeline & Experimental Design



- **Data:** 11 pairs, hourly, 2024–2026, 7 Majors & 4 EMs
- **94 features** → one common set per fold (correlation, MI, permutation importance)
- **Six M1 candidates:** RF, ExtraTrees, XGBoost, LightGBM, LSTM, GRU
- **M2 model:** RandomForest
- **Nested Optuna HPO:** models (outer), per-pair trading params (inner)

03

Results

Tournament: Choosing the Least Fragile Architecture

Mean across 11 pairs and 2 outer folds

	RF	ExtraTrees	XGBoost	LightGBM	LSTM	GRU
Sharpe Ratio ($\pm$ std)	0.72 ± 4.65	0.57 ± 4.87	1.97 ± 4.50	1.18 ± 4.49	1.27 ± 4.92	1.53 ± 5.12
Deflated Sharpe ($\% \pm$ std)	27.22 ± 39.04	25.56 ± 37.88	27.36 ± 37.45	32.6 ± 38.99	29.98 ± 36.27	25.42 ± 35.55
Accepted-signal accuracy ($\%$, fold 1 / 2)	57.2 / 68.7	56.5 / 72.8	58.3 / 74	57.5 / 70.3	56.8 / 63.6	58.6 / 68.7
Emp. FPR / ε (fold 1 / 2)	2.23 / 0.87	2.26 / 0.47	1.03 / 0.42	1.22 / 0.53	2.30 / 1.10	1.64 / 0.70

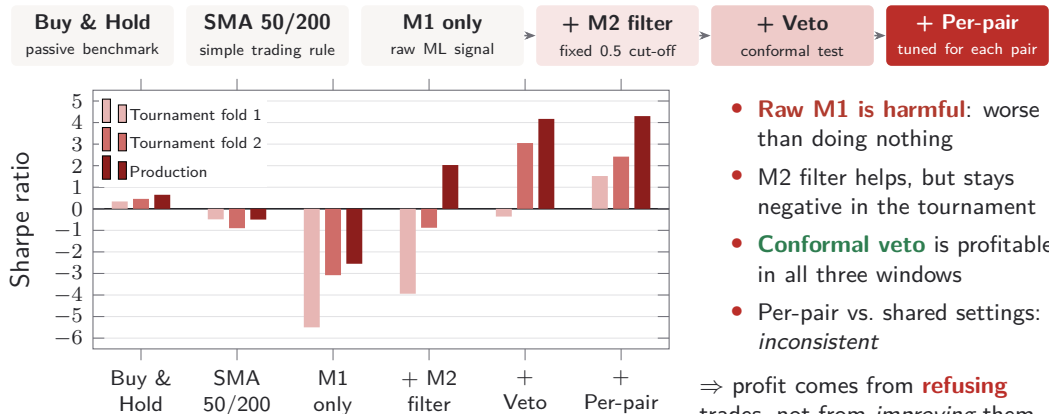
XGBoost

Reciprocal Rank Fusion
winner (score 0.19)

- Only XGBoost's veto stays close to its **calibrated** ε (ratio ≈ 1) and 4 of 6 let through $\sim 2\times$ the mistakes in fold 1
- Veto sharpens as the calibration memory grows (at the price of low recall)
- Std. $> 2\times$ mean: **no** architecture is uniformly profitable and the win is *relative*

What Actually Creates Value?

One building block at a time (XGBoost, avg. annualized Sharpe across 11 pairs)



Production: A very Concentrated Result

Global XGBoost refit, tested once on the untouched ~11-week window

Pair	Ret. %	Sharpe	DSR %	Trades
USDZAR	81.95	16.32	100.0	934
USDMXN	17.38	14.45	100.0	947
USDTRY	8.10	11.83	99.99	1019
USDCNH	1.67	3.13	39.56	730
EURUSD	0.59	3.86	54.45	8
USDJPY	-0.81	-2.30	0.31	184
5 majors	0.00	0.00	—	0

EMs above the line, majors below. DSR
undefined without trades

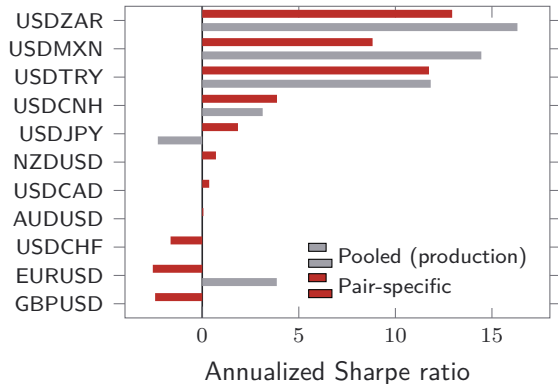
Majors' ε is a quantile of a **pooled**,
EM-dominated calibration set: **no potential
edge, or just a different score scale?**

Red flags

- Avg. Sharpe **4.30** beats both tournament folds (1.52, 2.42), annualized from 11 weeks
- **5 of 7 majors** never trade and ZAR, MXN, TRY give **90%** of summed Sharpes
- DSR < 50%: **USDCNH** and **USDJPY**; EURUSD borderline (54%, only 8 trades)
- DSR rules out search luck, *not* bad data: USDZAR may profit from **flash crashes** no other source confirms

Experiment: Pair-Specific Calibration

Only the calibration set changes: each pair vs. its *own* past mistakes



- **USDJPY** $-2.30 \rightarrow +1.85$: pooling *hid* a potential edge
- **GBP, EUR, CHF** now lose: pooling was an *accidental safeguard*
- USDCHF: same $\varepsilon = 0.02$, yet $0 \rightarrow 77$ trades
- Avg. Sharpe $4.30 \rightarrow 3.06$ ($\approx -30\%$): part of the production result is a *calibration artifact*

How real is any of it?

11 weeks is far too short to judge a trading strategy: a few lucky trades can produce a Sharpe of 4.3. These results show promise, not proof.

04

Conclusion

Answering the Five Objectives

	Objective	Verdict	Evidence
1	Architecture	Partial	XGBoost is the least fragile under a shared protocol, which is a comparative, not absolute, win
2	Meta-machinery	Yes	Raw M1 destroys value: the conformal veto is the load-bearing component as it <i>refuses</i> rather than <i>improves</i> trades
3	DSR	Partial	Flags USDCNH & USDJPY (EURUSD borderline), but is blind to regime, tail and data risk (USDZAR)
4	Features	Partial	Small stable core (VWMA, A/D, spread & volatility), whereas the rest reshuffles across folds
5	Global vs. local	Open	Per-pair vs. shared params (Kelly, SL/TP, ε , etc.): wins fold 1, loses fold 2. Per-pair ε decides <i>whether</i> a pair trades. Pooling hid USDJPY's edge <i>and</i> shielded losing majors

Conclusion & Outlook

- A **raw classifier cannot** drive a profitable intraday FX strategy
- Embedded in meta-labeling + conformal veto, it **can** be made profitable on a *subset* of pairs
- Profit comes from **principled abstention**, not better directional accuracy
- Remaining edge is concentrated, event-dependent and fragile: a *proof of concept* for the framework, not a deployable system

Next steps

- CPCV for more backtest paths
- A single institutional data feed
- Exogenous macro features
- Modular refactor of the codebase

Explore the Code

The full pipeline is open source:

- Data aggregation & sanity checks
- Feature engineering & triple-barrier labelling
- Architecture tournament, M1/M2 meta-labeling & conformal veto
- Custom backtester & all result figures

github.com/adamwetzli/master_thesis



Scan to open the repository



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Thank you

Questions & discussion

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